POLYTECHNIQUE MONTRÉAL
affiliée à l'Université de Montréal

A lightweight canonicalization engine in C

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Mémoire présenté en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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Ce mémoire intitulé :

A lightweight canonicalization engine in C

présenté par **William ZHANG**
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DEDICATION

À tous mes amis du labos,
vous me manquerez...

ACKNOWLEDGEMENTS

Texte / Text.

RÉSUMÉ

Le résumé est un bref exposé du sujet traité, des objectifs visés, des hypothèses émises, des méthodes expérimentales utilisées et de l'analyse des résultats obtenus. On y présente également les principales conclusions de la recherche ainsi que ses applications éventuelles. En général, un résumé ne dépasse pas trois pages.

Le résumé doit donner une idée exacte du contenu du mémoire ou de la thèse. Ce ne peut pas être une simple énumération des parties du manuscrit. Le but est de présenter de façon précise et concise la nature, l'envergure de la recherche, les sujets traités, les questions de recherche ou les hypothèses soulevées, les méthodes utilisées, les principaux résultats ainsi que les conclusions retenues. Un résumé ne doit jamais comporter de références ou de figures.

ABSTRACT

Written in English, the abstract is a brief summary similar to the previous section (Résumé). However, this section is not a word for word translation of the abstract in French.

The abstract is a brief statement of the subject matter, objectives, research questions or hypotheses, experimental methods and analysis of results. It also presents the main research conclusions and their possible applications. In general, an abstract should not exceed three pages.

The abstract should provide an exact idea of the thesis or dissertation's contents and it cannot be a simple enumeration of the manuscript's parts. The goal is to precisely and concisely present the nature and scope of the research. An abstract should never include references or figures. If the thesis or the dissertation is in English, the résumé (French-language abstract) should come first followed by the abstract.

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LIST OF SYMBOLS AND ACRONYMS

AD	Automatic Differentiation
AML	Algebraic Modeling Language
CSC	Compressed Sparse Column
CSR	Compressed Sparse Row
DAG	Directed Acyclic Graph
DCP	Disciplined Convex Programming
LP	Linear Program
MOI	MathOptInterface
MPS	Mathematical Programming System
NLP	Nonlinear Program
QP	Quadratic Program
SOCP	Second-Order Cone Program
SDP	Semidefinite Program

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CHAPTER 1 INTRODUCTION

Modeling languages for convex optimization such as CVXPY [1] and YALMIP [2] allow users to formulate optimization problems in a natural mathematical syntax. Through a process called *canonicalization*, these high-level problem descriptions are converted to a *standard conic form* used by many general-purpose solvers

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax + s = b \\ & s \in K, \end{array} \tag{1.1}$$

where $x \in \mathbf{R}^n$ and $s \in \mathbf{R}^m$ are the decision variables, and the *problem data* are the matrix $A \in \mathbf{R}^{m \times n}$, the vectors $b \in \mathbf{R}^m$ and $c \in \mathbf{R}^n$, and the non-empty closed convex cone $K \subseteq \mathbf{R}^m$. Almost all of optimization practice can be divided into two groups, modeling and solving. The modeler's role is to [...]. The solver's role is to find an optimal solution given a problem data instance. Modeling languages serve as the communication bridge between these two groups. When improvements in solvers are made, modeling languages must adapt to allow users to benefit from them. Conversely, it is through natural formulations that modelers can express their needs and help motivate solver improvements.

While the time taken to solve an optimization problem is often dominated by the solver, the time taken to *canonicalize* (or *compile*) a problem can also be significant, especially for large problems. In the case of CVXPY, the *compilation time* largely lies in *matrix stuffing*, the final step of canonicalization where symbolic expressions trees are converted to explicit numerical matrices. Other modeling languages take different approaches to form the problem data, we refer to their approaches as *coefficient extraction* and review them in §2.3.

Matrix stuffing. After expanding nonlinear functions into their *graph implementations* [3], all expressions trees are guaranteed to be affine. We review the entire canonicalization process and show a concrete example in §2.1. At this stage, the convex cone $\mathcal{K}$, which is typically a cartesian product of *standard cones*, is already computed and known. The remaining problem data are the *constraint matrix* A and the vectors b and c. The process to compute b is relatively straightforward, for example, one can *evaluate* the affine expression tree at the origin. The process to obtain c is a special case of the process to obtain A. For these reasons, we will often refer to matrix stuffing as the explicit task of forming A. We review matrix stuffing and its current implementations in CVXPY in §2.2.

Contributions. We make the following contributions:

1. We develop `SparseDiffEngine`.
2. We develop `SparseDiffPy`.
3. We benchmark our methods against existing approaches.

All code is open source and available at `https://github.com/SparseDifferentiation/`.

Thesis outline. The rest of this thesis is organized as follows. In §2 we review the literature on modeling languages, canonicalization, and matrix stuffing. In §3 we describe our proposed canonicalization engine. In §4 we present benchmark results comparing our method to existing approaches. Finally, in §5 we summarize our contributions and discuss future work.

CHAPTER 2 LITERATURE REVIEW

Solving an optimization problem on a computer requires translating the mathematical formulation into a data format that solvers can process. For linear programs, this means constructing the constraint matrix A, the right-hand side vector b, and the objective coefficients c that define the *standard form*,

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b, \\ & x \geq 0. \end{aligned} \tag{2.1}$$

Manual encoding In the early days of linear programming, analysts manually computed and encoded each coefficient onto punch cards. This process was tedious, error-prone and became impractical as linear programming found more and more applications in large-scale industrial problems [4].

MPS format IBM's Mathematical Programming System (MPS) file format, developed in the 1960s, established a standardized file format for representing linear programs [5]. The MPS format uses fixed-width fields designed for 80-column punch cards, with sections for rows, columns, and coefficients. William Orchard-Hays [6] provides a comprehensive history of these early mathematical programming systems, tracing their development from 1952 at RAND Corporation through their widespread adoption on mainframe computers in the late 1960s.

Matrix generators To automate the creation of MPS files, *matrix generators* emerged in the late 1960s; they are procedural programs that compute the problem coefficients through explicit loops and output the results in MPS format [7]. Commercial systems such as `MaGen` [8], `GAMMA` [9], and `DATAFORM` [10] were among the first matrix generators which were used in various domains.

2.1 Canonicalization

CVXPY. CVXPY is a domain-specific language for convex optimization embedded in Python, that was originally developed by Steven Diamond and Stephen Boyd at Stanford University [1]. Akshay Agrawal later redesigned the rewriting system in CVXPY [11], which

enabled support for other disciplined languages [12–14], and the project has since grown through contributions from many others.

References.

2.2 Matrix stuffing

Affine expression tree.

References.

2.3 Coefficient extraction

`JuMP` and `MathOptInterface`. JuMP is an algebraic modeling language embedded in Julia, originally developed in 2012 by Iain Dunning, Joey Huchette, and Miles Lubin at the MIT Operations Research Center [15, 16]. Its abstraction layer, MathOptInterface (MOI) [17], rethinks how a modeling language communicates with solvers, moving away from the traditional approach of reducing every problem to a single standard form.

Function-set pairs

`CasADi`. [18]

`AMPL`. [19]

References.

2.4 Computing derivatives

Finding the rate of change or the *derivative* of a function at a particular point is a fundamental concept in calculus that was discovered by Leibniz and Newton in the late 1600s. Since then, although the underlying theory has practically remained the same, various methods have been developed to efficiently compute derivatives in different contexts and applications. More recently, the rise of machine learning and deep learning brought renewed interest in this age-old problem through a new application of derivatives in training *neural networks*, it is known as the *backward pass* or *backpropagation* [20, 21]. Some theorists will claim

that backpropagation is little more than a clever application of the *chain rule* [22], but the large scale of modern workloads has led to the development of incredible software tools and implementation techniques which are interesting in their own right. This section provides an overview of these new tools and techniques, with a particular emphasis on *algorithmic differentiation* and its applications to computing sparse Jacobians [23].

Numerical differentiation. Other than computing derivatives by hand, the simplest method is to use *numerical differentiation*, which approximates the derivative of a function at a point by evaluating the function at nearby points using a *finite difference* formula similar to either

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \tag{2.2}$$

or

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \tag{2.3}$$

where h is a small step size. Although this approach is straightforward to implement and can achieve reasonable accuracy, there are important downsides to consider. First, it requires a linear number of function evaluations with respect to the number of input variables, which is expensive for high-dimensional functions. Second, it can easily lead to *truncation errors* (when h is too large) or *rounding errors* (when h is too small) in the computed derivative if the step size is not carefully chosen [24]. Despite these downsides, numerical differentiation is still widely used in practice, especially for verifying the correctness of derivative computations (also known as a *derivative checker*) [25].

Symbolic differentiation. Mathematical expressions can be manipulated automatically using simple derivative rules of calculus, such as the product and sum rule. This *symbolic* approach to differentiation is used in computer algebra systems such as Mathematica [26] and Maple [27] as well as the SymPy library in Python [28]. Unlike its numerical counterpart, symbolic differentiation tools compute derivatives as closed-form expressions, which can be simplified and evaluated at any arbitrary point. These closed-form expressions are particularly useful for deriving analytic solutions of optimization problems, and can be further used in mathematical proofs and analyses.

However, a naive implementation of symbolic differentiation can easily lead to exponentially large symbolic expressions, which is impractical and inefficient for many problems [23]. Interestingly, when intermediate expressions are stored as *common subexpressions*, and the expression is represented with a *directed acyclic graph* (DAG) instead of a tree, symbolic differentiation performs the same set of operations as *reverse mode differentiation* [29].

Algorithmic differentiation. There is a lot of confusion on the terminology surrounding the various methods for computing derivatives; *automatic differentiation* and *computational differentiation* are often used to describe the field, but we will instead refer to it as *algorithmic differentiation* since it emphasizes the algorithms behind the different implementations rather than a particular method (it is also the convention in the seminal book by Griewank and Walther [23]). Fundamentally, AD is a non-standard interpretation of a computer program: it augments the *evaluation trace* of a function (the sequence of floating point values calculated during the execution) with calculations of its derivatives [30]. This trace can be represented as a *computational graph*, a DAG in which each node corresponds to an elementary operation with known derivatives. Combining the derivatives of these operations throughout the graph using the chain rule gives the derivative of the overall function [23]. The computational graph is the central data structure of AD; the various *modes* of AD correspond to different ways elementary derivatives are accumulated on this graph.

Forward mode AD. The method of propagating tangents forward through the computational graph is known as *forward mode* AD. A single forward pass simultaneously computes the value and derivative of every intermediate variable with respect to a single input variable, giving the sensitivity of all outputs to that input. The cost of a forward pass is a small constant factor of the cost of evaluating the original function, since each elementary operation is augmented with at most a few additional arithmetic operations for the derivatives. However, one downside of forward mode AD is that the number of forward passes scales linearly with the number of input variables, which is impractical for certain functions (which have many input variables and very few output variables).

Reverse mode AD. In contrast to forward mode, *reverse mode* AD propagates adjoints backward through the computational graph. It first requires a forward pass to compute and store the value of every intermediate variable, and it then performs a reverse pass to compute the sensitivities of a single output variable with respect to all input variables. The cost of a reverse pass is also a small constant factor of the cost of evaluating the original function, but the number of passes required now scales linearly with the number of output variables rather than the number of input variables. This makes reverse mode AD particularly efficient for computing gradients of scalar-valued functions, making it the default method for modern machine learning libraries such as Jax [31], Pytorch [32], and Tensorflow [33].

Advantages of AD. [34].

Jacobian and Hessian computation.

Sparse Jacobian computation. graph coloring, cross-country elimination, vertex elimination, jacobian accumulation, etc. maybe mention graph coloring here?

[35] presents a code generator for operations on sparse data.

[36] presents a similar method for computing sparse Hessians with graph coloring. [37] is a survey of graph coloring methods for computing sparse Jacobians and Hessians.

Differentiation packages. mention ADOL-C and CppAD for C++. mention AMPL and CasADi for optimization. and also mention that they were covered above in the coeff extraction section.

References. The book by Griewank and Walther [23] is a comprehensive reference on the theory and practice of automatic differentiation. The book by Naumann [34] is another great reference that focuses on the implementation of AD in software, with a particular emphasis on the use of operator overloading and source code transformation techniques. [30] is an excellent and relatively up-to-date survey of the various methods for automatic differentiation in machine learning.

2.5 Sparse matrices

Matrices that contain many number of zero entries are known as *sparse matrices*. The simplest sparse matrix data structure is the *coordinate* or *triplet* format. It consists of two integer index arrays `rows` and `cols` and a floating point value array `vals` of length equal to the number of *nonzero entries* in the matrix.

For example, the matrix

$$\begin{bmatrix} 0 & 0 & 3.2 & 0 \\ 4.1 & 1.8 & 0 & 2.2 \\ 0 & 2.8 & 0 & 3.4 \end{bmatrix} \tag{2.4}$$

can be represented in coordinate format as

```
int    rows[] = {0,   1,    1,    1,    2,    2};
int    cols[] = {2,   0,    1,    3,    1,    3};
double vals[] = {3.2, 4.1, 1.8, 2.2, 2.8, 3.4};
```

where we use zero-based indexing and array declaration notation from the C programming language [38]. The coordinate format is the easiest to understand and form, but it is not

effective for most operations on sparse matrices; the *compressed sparse row* or CSR format is commonly used instead.

Compressed sparse row format. The CSR format differs from the coordinate format in that it replaces the `rows` array with a row pointer array `idxptr` of length equal to the number of rows plus one. The row pointer array indicates the starting index of each row in the `cols` arrays. For example, the matrix in (2.4) can be represented in CSR format as

```
int  idxptr[] = {0,  1,              4,        6};
int    cols[] = {2,  0,   1,   3,    1,   3};
double vals[] = {3.2, 4.1, 1.8, 2.2, 2.8, 3.4};
```

where we deliberately format `idxptr` to align with the start of each row in the `cols` arrays. The benefits of the CSR format are more than a reduced size of the `idxptr` array compared to the `rows` array. In the CSR format, accessing a row takes constant time and is very *cache-friendly* since the nonzero entries are stored contiguously in memory. For this reason, the CSR format is the default for *iterative methods* to solve sparse linear systems [39].

Compressed sparse column format. However, one downside of the CSR format is that accessing a column is much more difficult; one would need to look at every row to find the nonzero entries in a column. An alternative format is the *compressed sparse column* or CSC format, which swaps the role of the `idxptr` array to refer to column indices instead. In contrast to iterative methods, the CSC format is the preferred format for *direct methods* to solve sparse linear systems [40].

Other formats. The coordinate, CSR and CSC formats are the most widely used sparse matrix formats, but many others exist for different specialized use cases. The scientific Python sparse library `scipy.sparse` [41] allows users to form other types of sparse matrices such as the list of lists (LIL) matrix, the dictionary of keys (DOK) matrix, and the block-sparse row (BSR) matrix.

References. Although the work in this thesis does not necessarily revolve around solving sparse linear systems, a lot of techniques for manipulating sparse matrices we use were inspired by the literature. In particular, the book by Davis [40] was used extensively throughout the development of our methods. The book by Duff [42] is another great reference to learn more about the history and developement of sparse matrix algorithms. Another reference is the review article by the "gang of four" (Gill, Murray, Saunders and Wright) from the

Systems Optimization Laboratory at Stanford University [43] which provides a high-level overview of various applications of sparse matrix methods in optimization.

CHAPTER 3 A lightweight canonicalization engine

In this chapter we present a minimal implementation of a subset of affine atoms in the `SparseDiffEngine` package, and show how they can be used to compute the sparse constant Jacobian of a set of affine expressions. At this point, we are given a *canonicalized* problem in standard form, after the canonicalization process described in §2.1. The content of this chapter has many similarities to §2.2, since both serve the same purpose of forming the problem data. The structure of the chapter is as follows: in §3.1 we describe basic data structures in the engine which will be used throughout the atom implementations described in §3.2, and finally, in §3.3 we present a high level problem class interface which can easily be used in CVXPY. We also recommend reading §2.5 before reading this chapter, since it covers the basics of sparse matrices and their data structures.

3.1 Fundamental data structures

A CSR matrix struct in C looks like this.

```
typedef struct CSR_Matrix {
    int *p;      /* row pointers, size m+1 */
    int *i;      /* column indices, size nnz */
    double *x;   /* values, size nnz       */
    int m, n, nnz;
} CSR_Matrix;
```

The main data structure in the engine is the expression struct.

```
typedef struct expr {
    int   d1, d2, size; /* shape (d1 x d2) and size = d1*d2 */
    int   n_vars;       /* total number of variables     */
    int   var_id;       /* offset into u, or NOT_A_VARIABLE */

    struct expr *left;  /* left child (NULL for leaves)  */
    struct expr *right; /* right child (NULL for leaves) */

    double    *value;   /* forward-pass output, size entries */
    CSR_Matrix *jacobian; /* size x n_vars sparse Jacobian */

    forward_fn     forward;
```

```
    jacobian_init_fn jacobian_init_impl;
    eval_jacobian_fn eval_jacobian;
} expr;
```

every expression node has a shape (d1 x d2) and a size (d1*d2), and also two pointers to its left and right children. It will also have a pointer to its value, which is the output of the forward pass, and a pointer to its Jacobian, which is a CSR matrix. Every expression must also implement three functions: a forward function, a Jacobian initialization function, and a Jacobian evaluation function.

Leaf nodes. The two leaf nodes are variables and parameters. They are the fundamental building blocks of the expression tree.

```
expr *new_variable(int d1, int d2, int var_id, int n_vars)
{
    expr *node = (expr *) SP_CALLOC(1, sizeof(expr));
    init_expr(node, d1, d2, n_vars, forward,
            jacobian_init_impl, eval_jacobian, ...);
    node->var_id = var_id;

    return node;
}

expr *new_parameter(int d1, int d2, int param_id,
                    int n_vars, const double *values) {
    parameter_expr *pnode = SP_CALLOC(1, sizeof(parameter_expr));
    expr *node = &pnode->base;
    init_expr(node, d1, d2, n_vars, forward,
             jacobian_init_impl, eval_jacobian, ...);
    pnode->param_id = param_id;
    memcpy(node->value, values, node->size * sizeof(double));
    return node;
}
```

3.2 Atom implementations

We will now show the implementation of atoms in the engine. The semantics of an atom is as follows, it takes one or two expressions as input, and produces an output expression

corresponding to the result of the operation. Every atom implementation below will first show the basic CVXPY code, and then show how it is implemented in the engine.

Helpers and definitions. We first import the necessary libraries and define basic variables and constants which will be used throughout the implementation.

```
import cvxpy as cp
import numpy as np
import scipy.sparse as sp

x = cp.Variable(n1)
y = cp.Variable(n2)
a = cp.Parameter(n1)
b = cp.Parameter(n2)
```

Addition. The addition atom (also commonly referred to as `add`) takes two expressions and sums them together elementwise. The CVXPY code is as follows (we note that Python's operator overloading for `+` will automatically form an `add` atom).

```
add_expr = cp.add(x, y)
```

In the engine the implementation is as follows.

```
static void forward(expr *node, const double *u)
{
    /* children's forward passes */
    node->left->forward(node->left, u);
    node->right->forward(node->right, u);

    /* add left and right values */
    for (int i = 0; i < node->size; i++)
    {
        node->value[i] = node->left->value[i] + node->right->value[i];
    }
}

static void jacobian_init_impl(expr *node)
{
    /* initialize children's jacobians */
    jacobian_init(node->left);
```

```
    jacobian_init(node->right);

    /* we never have to store more than the sum of children's nnz */
    int nnz_max = node->left->jacobian->nnz + node->right->jacobian->nnz;
    node->jacobian = new_csr_matrix(node->size, node->n_vars, nnz_max);

    /* fill sparsity pattern */
    sum_csr_alloc(node->left->jacobian, node->right->jacobian, node->jacobian)
        ;
}

static void eval_jacobian(expr *node)
{
    /* evaluate children's jacobians */
    node->left->eval_jacobian(node->left);
    node->right->eval_jacobian(node->right);

    /* sum children's jacobians */
    sum_csr_fill_values(node->left->jacobian, node->right->jacobian, node->
        jacobian);
}


transpose_expr = cp.transpose(x)
```

Horizontal stack.

Vertical stack.

Indexing.

Sum.

Sum over an axis.

Elementwise multiplication.

Matrix multiplication.

3.3 Problem class and CVXPY interface

CHAPTER 4 Benchmark results

4.1 Benchmark Setup

4.1.1 Example

4.1.2 Runtime and Memory Usage

4.2 Results on Standard Benchmarks

4.3 Results on Stress Test Problems

4.3.1 Least Squares Problems

CHAPTER 5 CONCLUSION

Texte / Text.

5.1 Synthèse des travaux / Summary of Works

Texte / Text.

5.2 Limitations de la solution proposée / Limitations

5.3 Améliorations futures / Future Research

Texte / Text.

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APPENDIX A DÉMO

Texte de l'annexe A.